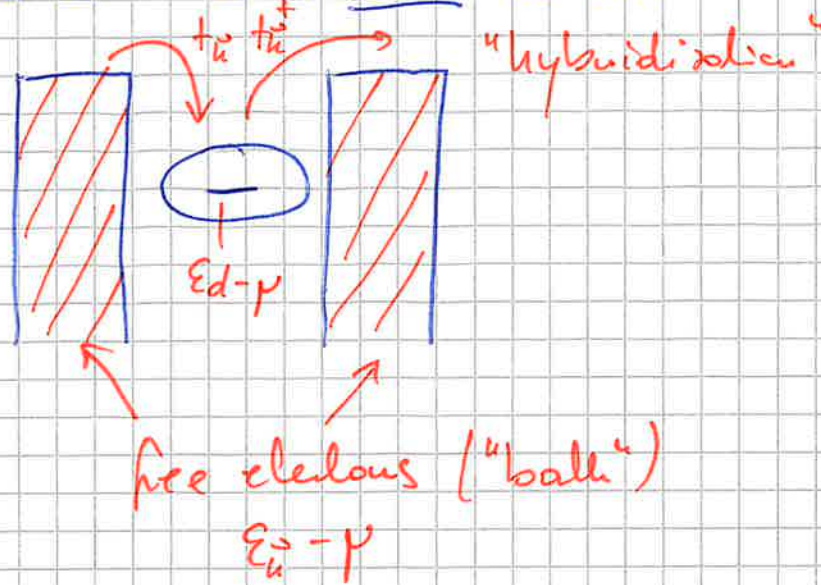


2.2 The Anderson impurity model

quantum dot with one atomic level ("d-electrons")



Taken together, we are left with the following Hamiltonian:

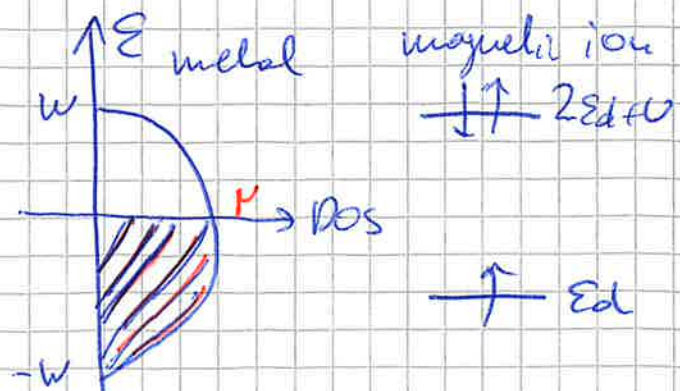
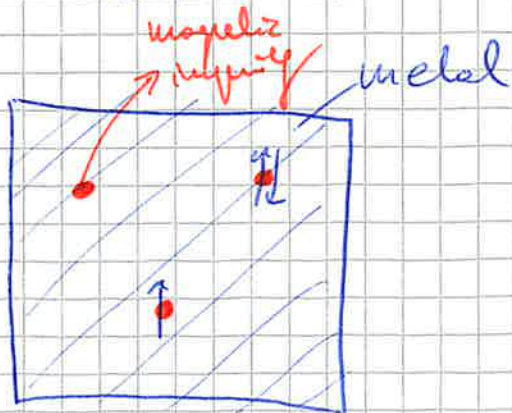
$$H = H_c + H_d + H_u + H_{\text{hyb}}$$

Conduction electrons: $\sum_{\vec{k}\sigma} (\epsilon_{\vec{k}} - \mu) c_{\vec{k}\sigma}^\dagger c_{\vec{k}\sigma}$

"impurity": $\sum_{\sigma} (\epsilon_d - \mu) c_{d\sigma}^\dagger c_{d\sigma} + U n_{d\uparrow} n_{d\downarrow}$

hybridization: $\sum_{\vec{k}\sigma} t_{\vec{k}} c_{d\sigma}^\dagger c_{\vec{k}\sigma} + \sum_{\vec{k}\sigma} t_{\vec{k}}^\dagger c_{\vec{k}\sigma}^\dagger c_{d\sigma}$

This Hamiltonian is called the **Anderson impurity model (AIM)**, because it was originally introduced to describe **magnetic impurities** embedded in a homogeneous host **metal**.



Although the AIM looks very simple, its solution, in fact, is very complicated and has a rich phase diagram!

E.g.: for certain system parameters, the model has a magnetic moment (~~maximum~~) (minimizing on-site interaction energy) or no magnetic moment (maximum hybridization energy)

Apart from the rich phase diagram, the AIM is a useful tool for (i) analyzing the so-called **Kondo problem** and (ii) the Hubbard model (within the so-called dynamical mean-field theory).

The first physical question which we want to answer is: when is the AIM magnetic?

For this analysis we will first introduce the **equation of motion theory** for the Green functions.

2.2.1 Equation of motion (EOM) theory for the single-particle Green function

retarded fermionic Green function:

$$G^R(\vec{r}, t; \vec{r}', t') = -i \theta(t - t') \langle \{ \psi(\vec{r}, t), \psi^\dagger(\vec{r}', t') \} \rangle$$

equation of motion: derivative w.r.t. first time argument

Let's take a general basis: $\{v\}$

$$H_0 = \sum_{vv'} t_{vv'} c_v^\dagger c_{v'}$$

$$G^R(vt, v't') = -i\theta(t-t') \langle \{c_v(t), c_{v'}^\dagger(t')\} \rangle$$

$$\begin{aligned} i\partial_t G^R(vt, v't') &= (-i)(i\partial_t \theta(t-t')) \langle \{c_v(t), c_{v'}^\dagger(t')\} \rangle + \\ &\quad + (-i)\theta(t-t') \langle \{i\partial_t c_v(t), c_{v'}^\dagger(t')\} \rangle = \\ &= \delta(t-t') \langle \{c_v(t), c_{v'}^\dagger(t)\} \rangle + \\ &\quad - i\theta(t-t') \langle \{i\partial_t c_v(t), c_{v'}^\dagger(t')\} \rangle = \\ &= \delta(t-t') \delta_{vv'} - i\theta(t-t') \langle \{i\partial_t c_v(t), c_{v'}^\dagger(t')\} \rangle \end{aligned}$$

What is the dependence on time of the equilibrium Green's function?

We assume that H is time-independent $\partial_t H = 0$

→ Heisenberg equation of motion

$$\begin{aligned} i\partial_t [c_v(t)] &= -[H, c_v(t)] = \\ &= -[H_0, c_v(t)] - [V_{\text{int}}, c_v(t)] \end{aligned}$$

quadratic part
(e.g. kinetic energy)

all interactions

$$\rightarrow -[H_0, c_v] = -\sum_{v'v''} t_{v'v''} [c_{v'}^\dagger c_{v''}, c_v]$$

Here, the following relation becomes useful:

$$[AB, C] = A[B, C] + [A, C]B = \\ = A\{B, C\} - \{A, C\}B$$

$$\Rightarrow [c_v^\dagger c_{v'}, c_v] = c_v^\dagger \underbrace{\{c_{v'}, c_v\}}_{\phi} - \underbrace{\{c_v^\dagger, c_v\}}_{\phi} c_{v'} \\ = \{c_v, c_v^\dagger\} = \delta_{vv'}$$

$$\Rightarrow -[H_0, c_v] = + \sum_{v'} t_{vv'} \delta_{vv'} c_{v'} = \sum_{v'} t_{vv'} c_{v'}$$

$$\Rightarrow \sum_{v'} (i\delta_{vv'} \partial_t - t_{vv'}) G^R(v'', v', t') = \delta(t-t') \delta_{vv'} + D^R(v'', v', t')$$

with $D^R(v'', v', t') = -i\theta(t-t') \langle [-[V_{int}, c_v](t), c_{v'}^\dagger] \rangle$

comparable to free-particle Green function

(in fact, they are higher-order Green functions)

Hamiltonian not time-dependent $\Rightarrow$ only differences $t-t'$
 $\Rightarrow$ Fourier transform

$$\sum_{v'} [\delta_{vv'} (i\omega + i\eta) - t_{vv'}] G^R(v'', v', \omega) = \delta_{vv'} + D^R(v'', v', \omega)$$

$$D^R(v'', v', \omega) = -i \int_{-\infty}^{\infty} dt e^{i(\omega + i\eta)(t-t')} \theta(t-t') \langle [-[V_{int}, c_v](t), c_{v'}^\dagger(t')] \rangle$$

Examples for the EOM technique:

•) non-interacting particles

$D^R = 0 \Rightarrow$ solve equation

$$\sum_{v''} (S_{vv''}(\omega + i\eta) - t_{vv''}) G_0^R(v''v', \omega) = S_{vv'}$$

inverse Green function: $(G_0^R)^{-1}(vv', \omega) = S_{vv'}(\omega + i\eta) - t_{vv'}$

$$= \underline{(G_0^R)^{-1}}_{vv'} \quad \text{matrix}$$

$\rightarrow \underline{(G_0^R)^{-1}} \underline{G_0^R} = \mathbb{1} \Rightarrow$ invert equation for finding Green function

diagonal basis: $t_{vv'} = S_{vv'} \epsilon_v$

$$\underline{(G_0^R)^{-1}}_{vv'} = G_0^R(v, \omega) S_{vv'} = \frac{1}{\omega - \epsilon_v + i\eta} S_{vv'}$$

•) Single level coupled to continuum

$$H = H_0 + H_{\text{hyb}} + H_e \Rightarrow \text{no interactions, } V_{\text{int}} = 0$$

non-
int.
electron gas

hybridization

atomic level

$$H_0 = \sum_v \epsilon_v c_v^\dagger c_v, \quad \epsilon_v = \epsilon_v - \mu$$

$$H_e = \epsilon_0 c_e^\dagger c_e, \quad H_{\text{hyb}} = \sum_v (t_{v\uparrow}^\dagger c_v^\dagger c_e + t_{v\downarrow} c_e^\dagger c_v)$$

We are interested in finding the dressed Green function for the **local state**. For this we need two Green functions:

$$G^R(l, l, t-t') = -i\theta(t-t') \langle \{c_l(t), c_l^\dagger(t')\} \rangle$$

$$G^R(v, l, t-t') = -i\theta(t-t') \langle \{c_v(t), c_l^\dagger(t')\} \rangle$$

for fermions (electrons).

We let v run over both the local state l and all continuum states (denoted by v):

$$\sum_{v''} [\delta_{vv''}(\omega + i\gamma) - t_{vv''}] G^R(v'', v'; \omega) = \delta_{vv'}$$

We need the cases ① $v=v'=l$

② $v \in \text{continuum}, v'=l$

$$\textcircled{1} \sum_{v'' \in \{l, \text{continuum}\}} [\delta_{lv''}(\omega + i\gamma) - t_{lv''}] G^R(v'', l; \omega) = 1$$

$$(\omega + i\gamma - \epsilon_0) G^R(l, l; \omega) - \sum_v t_v G^R(v, l; \omega) = 1$$

$$\textcircled{2} \sum_{v'' \in \{l, \text{continuum}\}} [\delta_{vv''}(\omega + i\gamma) - t_{vv''}] G^R(v'', l; \omega) = 0$$

$$(\omega + i\gamma - \epsilon_v) G^R(v, l; \omega) - t_v^* G^R(l, l; \omega) = 0$$

$$\Rightarrow G^R(v, l; \omega) = \frac{t_v^* G^R(l, l; \omega)}{\omega - \epsilon_v + i\gamma}$$

insert in ①

$$\Rightarrow G^R(l, l; \omega) = \frac{1}{\omega - \epsilon_0 - \sum_v \frac{|H_{lv}|^2}{\omega - \epsilon_v + i\eta}} =$$

$$= \frac{1}{\omega - \epsilon_0 - \Sigma^R(\omega)} \quad \textcircled{4}$$

Although we treat a non-interacting system, we can define a "self-energy" for the single-particle level. It changes the pole structure of the Green function and adds a lifetime to the impurity state.

It arises out of **off-diagonal terms** in the Hamiltonian
 $\Rightarrow$ no large diagonal in the l -space
 $\Rightarrow$ the diagonal state is superposition of v and l !

⑧ compare this to the Green function of an atom hybridized with a flat conduction bath of electrons (first MMS tutorial)!

